



JKBOSE 2026

CLASS 11TH

SEQUENCES AND SERIES

MATHS

LECTURE-04

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TOPICS TO BE COVERED

1 Relationship between AM and GM (AM- GM inequality)

2 Related Problems

$$A = \frac{a+b}{2}$$

$$G = \sqrt{ab}$$

3 Exercise Discussion 8.2

Relationship Between A.M. and G.M

We know that for any two positive numbers a & b ,

$$(\sqrt{a} - \sqrt{b})^2 \geq 0 \text{ with equality } a=b$$

$$\Rightarrow \cancel{\sqrt{a}^2} + \cancel{\sqrt{b}^2} - 2\sqrt{ab} \geq 0$$

$$\Rightarrow a+b \geq 2\sqrt{ab}$$

$$\Rightarrow \left(\frac{a+b}{2}\right) \geq \sqrt{ab}$$

$$\Rightarrow A \geq G, \text{ with equality } a=b.$$

Arithmetic Mean: let $x_1, x_2, x_3, \dots, x_n$ be any n -positive numbers, then their A.M. is denoted & defined as

$$A = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

Geometric Mean: let $x_1, x_2, x_3, \dots, x_n$ be any n -positive numbers, then their G.M is denoted & defined as:

$$G = (x_1 x_2 x_3 \dots x_n)^{\frac{1}{n}}$$

$$* \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \geq (x_1 x_2 \dots x_n)^{\frac{1}{n}} \text{ with } x_1 = x_2 = x_3 = \dots = x_n$$

EXAMPLE

If A.M. and G.M. of two positive numbers a and b are 10 and 8, respectively, find the numbers.

(Sol) It's given that $A = 10$ & $G = 8$

$$\Rightarrow \left(\frac{a+b}{2}\right) = 10 \Rightarrow a = 20 - b \quad \text{--- (1)}$$

$$\text{Also, } \sqrt{ab} = 8 \Rightarrow (\sqrt{ab})^2 = 8^2$$

$$\Rightarrow ab = 64 \quad \text{--- (2)}$$

Using (1) in (2), we get;

$$(20 - b)b = 64$$

$$\Rightarrow 20b - b^2 = 64$$

$$\Rightarrow b^2 - 20b + 64 = 0$$

$$\Rightarrow (b - 16)(b - 4) = 0$$

$$\Rightarrow b = 16, 4$$

$$\text{So, } a = 4, 16$$

Therefore the required numbers are 4 & 16.

Hence the result.

QUESTION (NCERT EXAMPLAR)

The minimum value of the expression $3^x + 3^{1-x}$, $x \in \mathbf{R}$, is

- (Sol) Use AM-GM inequality
Here $a = 3^x$, $b = 3^{1-x}$.
- 1 ☒ 0 ☒ $\times$
- 2 ☒ $1/3$ ☒ $\times$ So, $A \geq G$
 $\Rightarrow \left(\frac{3^x + 3^{1-x}}{2} \right) \geq \sqrt{3^x \times 3^{1-x}}$
- 3 ☒ 3 ☒ $\times$ $\Rightarrow 3^x + 3^{1-x} \geq 2\sqrt{3}$ with equality $x = \frac{1}{2}$
- 4 ☒ $2\sqrt{3}$ ☒ $\checkmark$ $\Rightarrow$ Minimum value of $3^x + 3^{1-x} = 2\sqrt{3}$.

QUESTION (NCERT EXAMPLAR)

The minimum value of $4^x + 4^{1-x}$, $x \in \mathbf{R}$, is

- 1 2 ✗ $\left(\frac{4^x + 4^{1-x}}{2} \right) \geq \sqrt{4^x \times 4^{1-x}}$
- 2 4 ✓ $\Rightarrow \left(\frac{4^x + 4^{1-x}}{2} \right) \geq 2$
- 3 1 ✗ $\Rightarrow 4^x + 4^{1-x} \geq 4$ with equality $x = \frac{1}{2}$
- 4 0 ✗

QUESTION

The minimum value of $\frac{x^4 + 8}{x^2}$ is:

- 1 $2\sqrt{2}$ ✗
- 2 $4\sqrt{8}$ ✗
- 3 $4\sqrt{2}$ ✓
- 4 None of these ✗

(Sol) we have;

$$\frac{x^4 + 8}{x^2} = \frac{x^4}{x^2} + \frac{8}{x^2}$$

$$\Rightarrow \frac{x^4 + 8}{x^2} = x^2 + \frac{8}{x^2} \geq 2\sqrt{x^2 \times \frac{8}{x^2}}$$

$$\Rightarrow \frac{x^4 + 8}{x^2} \geq 2 \times 2\sqrt{2}$$

$$\Rightarrow \boxed{\frac{x^4 + 8}{x^2} \geq 4\sqrt{2}} \text{ with equality } x = 2^{3/4}$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

$$a+b \geq 2\sqrt{ab}$$

$$\frac{8}{x^2} = x^2$$

$$x^4 = 8$$

$$x = (8)^{1/4} = 2^{3/4}$$

QUESTION (BLACK BOOK PROBLEM)

If the equation $x^4 - 4x^3 + ax^2 + bx + 1 = 0$ has four positive roots, then the value of $(a + b)$ is :

1 -4 ✗

2 2 ✓

3 6 ✗

4 Cannot be determined ✗

(Sol) let x_1, x_2, x_3, x_4 be the four positive roots of the given equation. So,

$$x_1 + x_2 + x_3 + x_4 = -(-4)$$

$$\Rightarrow x_1 + x_2 + x_3 + x_4 = 4 \quad \& \quad x_1 x_2 x_3 x_4 = 1.$$

$$\Rightarrow \frac{x_1 + x_2 + x_3 + x_4}{4} = 1 \quad \& \quad (x_1 x_2 x_3 x_4)^{\frac{1}{4}} = (1)^{\frac{1}{4}} = 1.$$

$$A = 1, G = 1$$

$$\Rightarrow A = G$$

$$\Rightarrow x_1 = x_2 = x_3 = x_4 = 1$$

$$\Rightarrow \text{So, } (1)^4 - 4(1)^3 + a(1)^2 + b(1) + 1 = 0$$

$$\Rightarrow 1 - 4 + a + b + 1 = 0$$

$$\Rightarrow 2 - 4 + a + b = 0.$$

$$\Rightarrow a + b = 2$$

QUESTION (JEE MAINS 2020)

If $2^{1-x} + 2^{1+x}, f(x), 3^x + 3^{-x}$ are in A.P, then the minimum value of $f(x)$ is:

1

1 X

2

2 X

3

3 ✓

4

4 X

(Sol) Since $2^{1-x} + 2^{1+x}, f(x), 3^x + 3^{-x}$ are in A.P.

$$\text{So, } f(x) - (2^{1-x} + 2^{1+x}) = (3^x + 3^{-x}) - f(x)$$

$$\Rightarrow 2f(x) = (2^{1-x} + 2^{1+x}) + (3^x + 3^{-x})$$

$$\Rightarrow f(x) = \left(\frac{2^{1-x} + 2^{1+x}}{2} \right) + \left(\frac{3^x + 3^{-x}}{2} \right)$$

$$\geq \sqrt{2^{1-x} \times 2^{1+x}} + \sqrt{3^x \times 3^{-x}}, \text{ with equality at } x=0.$$

$$= 2 + 1 = 3$$

$$\Rightarrow f(x) \geq 3$$

$$\Rightarrow f(x) \Big|_{\min} = 3.$$

QUESTION (JEE MAINS 2019)

The number of real solutions of the equation $\sin(e^x) = 2^x + 2^{-x}$ is:

1 0 ✓

2 1 ✗

3 2 ✗

4 Infinitely many ✗

$$\begin{aligned} \text{(Sol)} \quad \left(\frac{2^x + 2^{-x}}{2} \right) &\geq \sqrt{2^x \times 2^{-x}} \quad -1 \leq \sin(e^x) \leq 1 \quad \geq 2 \\ &\Rightarrow 2^x + 2^{-x} \geq 2 \end{aligned}$$

EXAMPLE

If $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ is the A.M. between a and b . Then, find the value of n .

$$a^x = a^y \Rightarrow x = y, a \neq 0, 1, -1$$

(sol) Here $\frac{a^{n+1} + b^{n+1}}{a^n + b^n} = \frac{a+b}{2}$

$$\Rightarrow 2a^{n+1} + 2b^{n+1} = (a+b)(a^n + b^n)$$

$$\Rightarrow 2a^{n+1} + 2b^{n+1} = a^{n+1} + ab^n + ba^n + b^{n+1}$$

$$\Rightarrow a^{n+1} - ba^n = ab^n - b^{n+1}$$

$$\Rightarrow a^n \cancel{(a-b)} = b^n \cancel{(a-b)} (a \neq b)$$

$$a^n = b^n \Rightarrow \left(\frac{a}{b}\right)^n = 1 \Rightarrow \left(\frac{a}{b}\right)^n = \left(\frac{a}{b}\right)^0$$

$$\frac{a}{b} \neq 0, -1, 1$$

$$\Rightarrow n = 0$$

QUESTION (NDA-2016)

What is the greatest value of the Positive integer n satisfying the condition?

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots + \frac{1}{2^{n-1}} < 2 - \frac{1}{1000}$$

- 1** 8
- 2** 9
- 3** 10
- 4** 11

HP/W

QUESTION (NDA-2025)

If $(10 + \log_{10} x)$, $(10 + \log_{10} y)$ and $(10 + \log_{10} z)$ are in A.P., then consider the following statements :

I. The GM of x and z is y^2

II. The AM of $\log_{10} x$ and $\log_{10} z$ is $\log_{10} y$.

Which of the statements given above is/are correct ?

1 I only

HP/W

2 II only

3 Both I and II

4 Neither I and II

QUESTION (NDA-2025)

If $p^x = q^y = r^z$, where x, y and z are in GP, then consider the following statements

I. p, q and r are in AP

II. $\ln p, \ln q$ and $\ln r$ are in GP

Which of the statements given above is/are correct ?

1 I only

HP/w

2 II only

3 Both I and II

4 Neither I and II

EXERCISE – 8.2

QUESTION

Find the 20^{th} and n^{th} terms of the G.P. $\frac{5}{2}, \frac{5}{4}, \frac{5}{8}, \dots$

(Sol) Given G.P. : $\frac{5}{2}, \frac{5}{4}, \frac{5}{8}, \dots$

$$\text{Here } a = \frac{5}{2}, r = \frac{1}{2}.$$

$$\text{So } a_n = ar^{n-1} = \frac{5}{2} \times \frac{1}{2^{n-1}}$$

$$\Rightarrow a_n = \frac{5}{2^n}$$

$$\text{Also, } a_{20} = \frac{5}{2^{20}}.$$

QUESTION

Find the 12th term of a G.P. whose 8th term is 192 and the common ratio is 2.

(Sol) Given $a_8 = 192$ & $r = 2$

$$\Rightarrow ar^7 = 192.$$

$$\Rightarrow a(2)^7 = 192.$$

$$\Rightarrow a \times 128 = 192$$

$$\Rightarrow a = \frac{192}{128} = \frac{6}{4} = \frac{3}{2}$$

$$\Rightarrow \text{So } a = \frac{3}{2}, r = 2$$

$$\Rightarrow a_{12} = ar^{11} = \frac{3}{2} \times 2^{11}$$

$$\Rightarrow a_{12} = 3 \times 2^{10}$$

QUESTION

The 5th, 8th and 11th terms of a G.P. are p , q and s , respectively. Show that $q^2 = ps$.

Proof: Given that $a_5 = p$
 $a_8 = q$
 $a_{11} = s$

$$\Rightarrow ar^4 = p, ar^7 = q, ar^{10} = s.$$

$$\Rightarrow ps = ar^4 \times ar^{10} = a^2 r^{14}$$

$$\& q^2 = (ar^7)^2 = a^2 r^{14} = ps$$

$$q^2 = ps$$

Hence proved.

QUESTION

The 4th term of a G.P. is square of its second term, and the first term is - 3.

Determine its 7th term.

$$a_4 = a_2^2, a = -3$$

$$ar^3 = (ar)^2$$

$$\Rightarrow \cancel{a}r^3 = a^{\cancel{2}} \times r^2$$

$$\Rightarrow \cancel{r}^2 \times r = a \times \cancel{r}^2$$

$$\Rightarrow r = a$$

QUESTION



Which term of the following sequences:

(a) ^{H/W} $2, 2\sqrt{2}, 4, \dots$ is 128?

~~(b)~~ $\sqrt{3}, 3, 3\sqrt{3}, \dots$ is 729?

(c) ^{H/W} $\frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots$ is $\frac{1}{19683}$?

(Sol) (b) let $a_n = 729$

$$\Rightarrow ar^{n-1} = 729$$

$$\Rightarrow \sqrt{3} \times \sqrt{3}^{n-1} = 3^6$$

$$\Rightarrow \sqrt{3}^n = 3^6$$

$$\Rightarrow (3)^{\frac{n}{2}} = 3^6$$

$$\Rightarrow \frac{n}{2} = 6 \Rightarrow n = 12$$

$$\frac{81 \times 3}{243}$$

$$3^4 = 81$$

$$3^5 = 243$$

$$3^6 = 243 \times 3$$

$$= 729$$

QUESTION

For what values of x , the numbers $-\frac{2}{7}, x, -\frac{7}{2}$ are in G.P.?

H/W

$$\frac{x}{\left(-\frac{2}{7}\right)} = \frac{\left(-\frac{7}{2}\right)}{x}$$

$$\Rightarrow x^2 = \cancel{-\frac{7}{2}} \times \cancel{-\frac{2}{7}}$$

$$x^2 = 1$$

$$x = \pm 1.$$

QUESTION

Find the sum of indicated number of terms in each of the geometric progressions in Exercises 7 to 10:

0.15, 0.015, 0.0015, ... 20 terms.

(Sol) let $S_{20} = 0.15 + 0.015 + 0.0015 + \dots$ upto 20 terms

$$\Rightarrow S_{20} = \frac{15}{100} + \frac{15}{1000} + \frac{15}{10000} + \dots \text{upto 20 terms}$$

$$\Rightarrow S_{20} = \frac{15}{10^2} + \frac{15}{10^3} + \frac{15}{10^4} + \dots + \frac{15}{10^{21}}$$

$$= 15 \left(\frac{1}{10^2} + \frac{1}{10^3} + \dots + \frac{1}{10^{21}} \right)$$

$$\Rightarrow S_{20} = 15 \left(\frac{\frac{1}{10^2} \left(1 - \frac{1}{10^{20}} \right)}{\left(1 - \frac{1}{10} \right)} \right)$$

$$\Rightarrow S_{20} = 15 \left\{ \left(\frac{1}{10^2} - \frac{1}{10^{22}} \right) \times \frac{10}{9} \right\}$$

$$\Rightarrow S_{20} = \cancel{15} \left\{ \left(\frac{1}{10} - \frac{1}{10^{21}} \right) \times \frac{1}{\cancel{3}\cancel{9}} \right\}$$

$$\Rightarrow S_{20} = \frac{5}{3} \left(\frac{1}{10} - \frac{1}{10^{21}} \right)$$

QUESTION

Find the sum of indicated number of terms in each of the geometric progressions in Exercises 7 to 10:

$\sqrt{7}, \sqrt{21}, 3\sqrt{7}, \dots n$ terms.

$$S_n = \frac{a(r^n - 1)}{r - 1}, r \neq 1, r > 1$$

$$\frac{a(r^n - 1)}{r - 1}, r > 1$$

$$\frac{a(1 - r^n)}{1 - r}, r < 1$$

QUESTION

Find the sum of indicated number of terms in each of the geometric progressions in Exercises 7 to 10:

$1 - a, a^2, -a^3, \dots n$ terms (if $a \neq -1$).

$$S_n = \frac{a(r^n - 1)}{r - 1}, r \neq 1.$$

H/W

QUESTION

Find the sum of indicated number of terms in each of the geometric progressions in Exercises 7 to 10:

$$x^3, x^5, x^7, \dots n \text{ terms (if } x \neq \pm 1)$$

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QUESTION

Evaluate $\sum_{k=1}^{11} (2 + 3^k)$.

(Sol) We have ;

$$\begin{aligned}\sum_{k=1}^{11} (2 + 3^k) &= \sum_{k=1}^{11} 2 + \sum_{k=1}^{11} 3^k \\ &= 22 + (3 + 3^2 + 3^3 + \dots + 3^{11}) \\ &= 22 + \frac{3(3^{10} - 1)}{3 - 1} = 22 + \frac{3(3^{10} - 1)}{2}.\end{aligned}$$

QUESTION

The sum of first three terms of a G.P. is $\frac{39}{10}$ and their product is 1. Find the common ratio and the terms.

H/w.

$$\left[\frac{a}{r}, a, ar \right]$$

QUESTION

7, 77, 777, ...



Find the sum to n terms of the sequence, 8, 88, 888, 8888.....

H/w

QUESTION

Insert two numbers between 3 and 81 so that the resulting sequence is G.P.

H/W

4 hours ✓
Marathon
100m

Thank
You